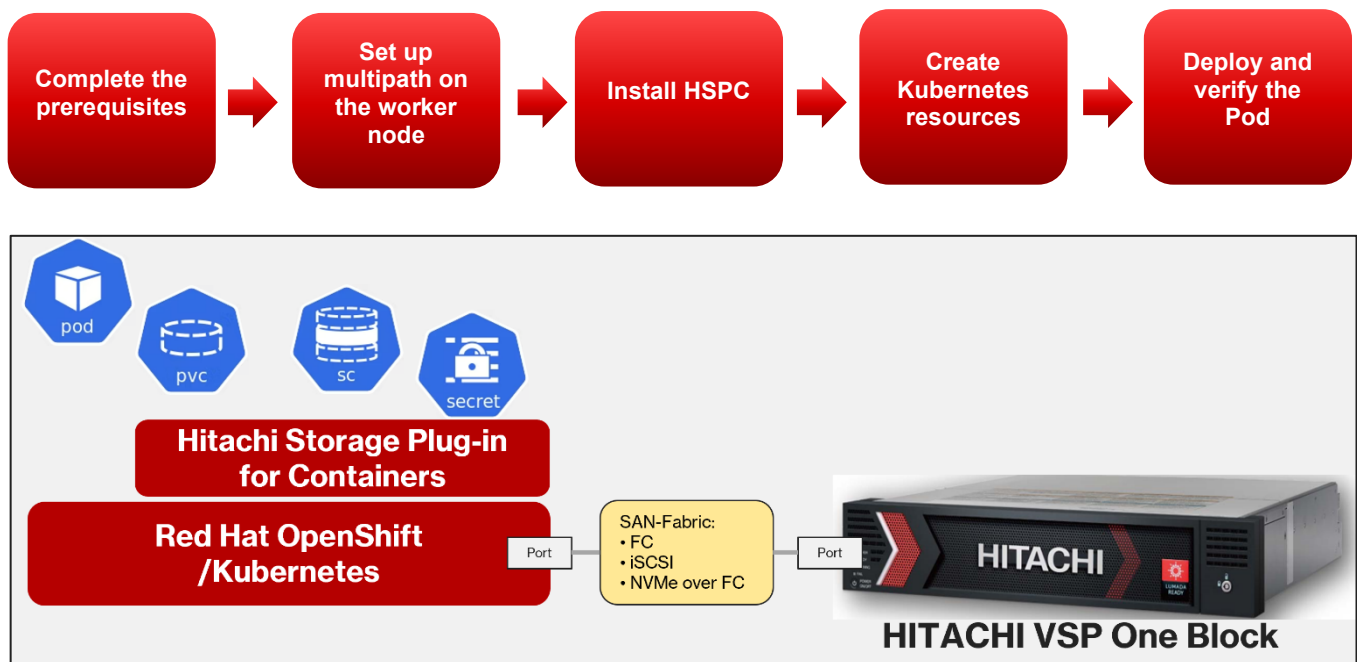


Provision Storage Volumes for Red Hat OpenShift Container Platform with Hitachi Storage Plug-in for Containers

Hitachi Storage Plug-in for Containers (HSPC) is a block Container Storage Interface (CSI) driver designed to create and manage persistent volumes for Hitachi storage systems in a Kubernetes container platform.

With HSPC, you can create and run stateful containers or pods supported by resilient persistent storage. It facilitates the dynamic provisioning of persistent volumes from Hitachi Virtual Storage Platform One (VSP One) storage systems.

Get started with these simple steps to provision storage volumes for containers.



Step 1: Complete the prerequisites

In VSP One Block Administrator, make sure of the following:

1. The storage pool is configured for Hitachi Dynamic Provisioning (HDP).
2. The storage port is configured for the following settings:
 - Connection (Fibre Channel): P-to-P (Point-to-Point)
 - Fabric: ON
 - Security: Enabled

Step 2: Set up multipath on the worker nodes

Set up multipath on the worker nodes, which are part of the Red Hat® OpenShift® environment to ensure high availability of the volume. At least two worker nodes are required. You can add more nodes based on your requirements.

1. Obtain the `multipath-machineconfig-sample.yaml` file from the following URL:

<https://github.com/hitachi-vantara/csi-operator-hitachi>

2. Apply the multipath settings on each worker node:

```
#oc apply -f multipath-machineconfig-sample.yaml
```

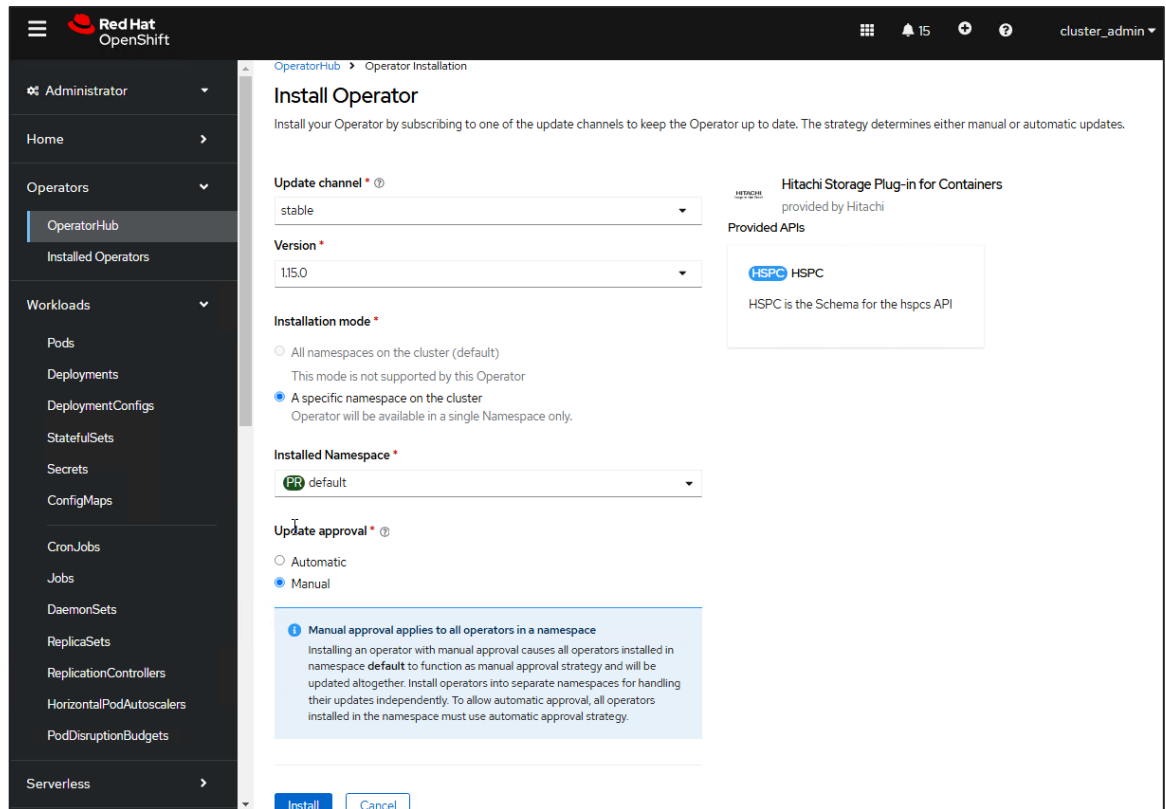
Step 3: Install HSPC

You can deploy HSPC to OpenShift using the Operator. Install the Operator from the OperatorHub in the OpenShift web console.

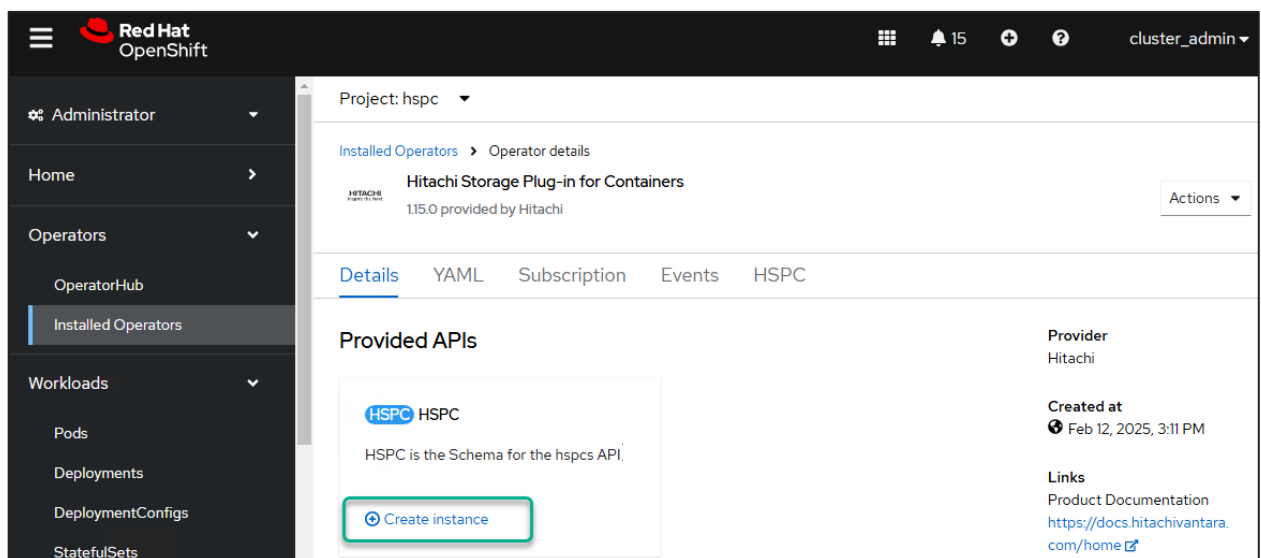
1. Log in to the Red Hat OpenShift web console.
 - a. Navigate to **Operators > OperatorHub**.

b. Select the following settings, and then click **Install**.

- Installation mode: Select **A specific namespace on the cluster** and specify a namespace.
- Update approval: Select **Manual**.



2. In the Navigation pane, click **Installed Operators**, and then click **Create instance**.



3. Confirm that the status **READY** is **true**:

```
#oc get hspc -n <Storage-Plug-in-for-Containers-namespace>

NAME    READY    AGE
hspc    true     30s
```

Step 4: Create Kubernetes resources

Create a Secret and a StorageClass for Kubernetes.

1. Create a Secret and set the user and password to access the relevant asset in the storage system from HSPC.

- a. Create a Secret yaml file:

```
apiVersion: v1
kind: Secret
metadata:
  name: secret-sample
type: Opaque
data:
  # base64 encoded storage url. E.g.: echo -n "http://192.0.2.1" | base64
  url: MTkyLjAuMi4x=
  # base64 encoded storage username. E.g.: echo -n "User01" | base64
  user: VXNlcjAx
  # base64 encoded storage password. E.g.: echo -n "Password01" | base64
  password: UGFzc3dvcmQwMQ==
```

Key	Description
name	Enter a name for the Secret.
url	Enter your storage system URL in Base64.
user	Enter your storage system username in Base64.
password	Enter your storage system password in Base64.

If no namespace is specified, it will be created in the default or the current namespace.

- b. Create a Secret:

```
oc create -f <yaml-file-name-of-the-secret-file>
```

2. Create a StorageClass. The StorageClass of the Kubernetes resource specifies the storage, port, and interface for provisioning volumes on the storage system from HSPC.

- a. Log in to VSP One Block Administrator and obtain the following information:

- Serial number
- Pool ID
- Port ID (Number of worker nodes to use)

- b. Create a StorageClass yaml file:

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: sc-sample
  annotations:
    kubernetes.io/description: Hitachi Storage Plug-in for Containers
provisioner: hspc.csi.hitachi.com
reclaimPolicy: Delete
volumeBindingMode: Immediate
allowVolumeExpansion: true
parameters:
  serialNumber: "54321"
  poolID: "1"
  portID: CL1-A,CL2-A
  connectionType: fc
  csi.storage.k8s.io/fstype: ext4
  csi.storage.k8s.io/node-publish-secret-name: "secret-sample"
  csi.storage.k8s.io/node-publish-secret-namespace: "default"
  csi.storage.k8s.io/provisioner-secret-name: "secret-sample"
  csi.storage.k8s.io/provisioner-secret-namespace: "default"
  csi.storage.k8s.io/controller-publish-secret-name: "secret-sample"
  csi.storage.k8s.io/controller-publish-secret-namespace: "default"
  csi.storage.k8s.io/node-stage-secret-name: "secret-sample"
  csi.storage.k8s.io/node-stage-secret-namespace: "default"
```

```
csi.storage.k8s.io/controller-expand-secret-name: "secret-sample"
csi.storage.k8s.io/controller-expand-secret-namespace: "default"
```

Key	Description
name	Enter a name for the StorageClass.
serialNumber	Enter the serial number you obtained in step 2a.
poolID	Enter the pool ID you obtained in step 2a.
portID	Enter the port ID you obtained in step 2a.
secret-name	Change the secret name to the name of the Secret file you created.
secret-namespace	Enter a name for the secret namespace.

- c. Create the StorageClass:

```
oc create -f <name-of-the-yaml-file-for-the-storage-class>
```

Step 5: Deploy and verify the Pod

To dynamically create and mount persistent volumes into pods, specify a PersistentVolumeClaim (PVC) with the required storage volume characteristics in the pod.

1. Create a PVC. A volume and a PVC have a one-to-one relationship. To create multiple volumes, create multiple PVCs.

The PVC Kubernetes resource defines the volume requirements for the pod, including capacity and access mode.

- a. Create a PVC yaml file:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: pvc-sample
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: sc-sample
```

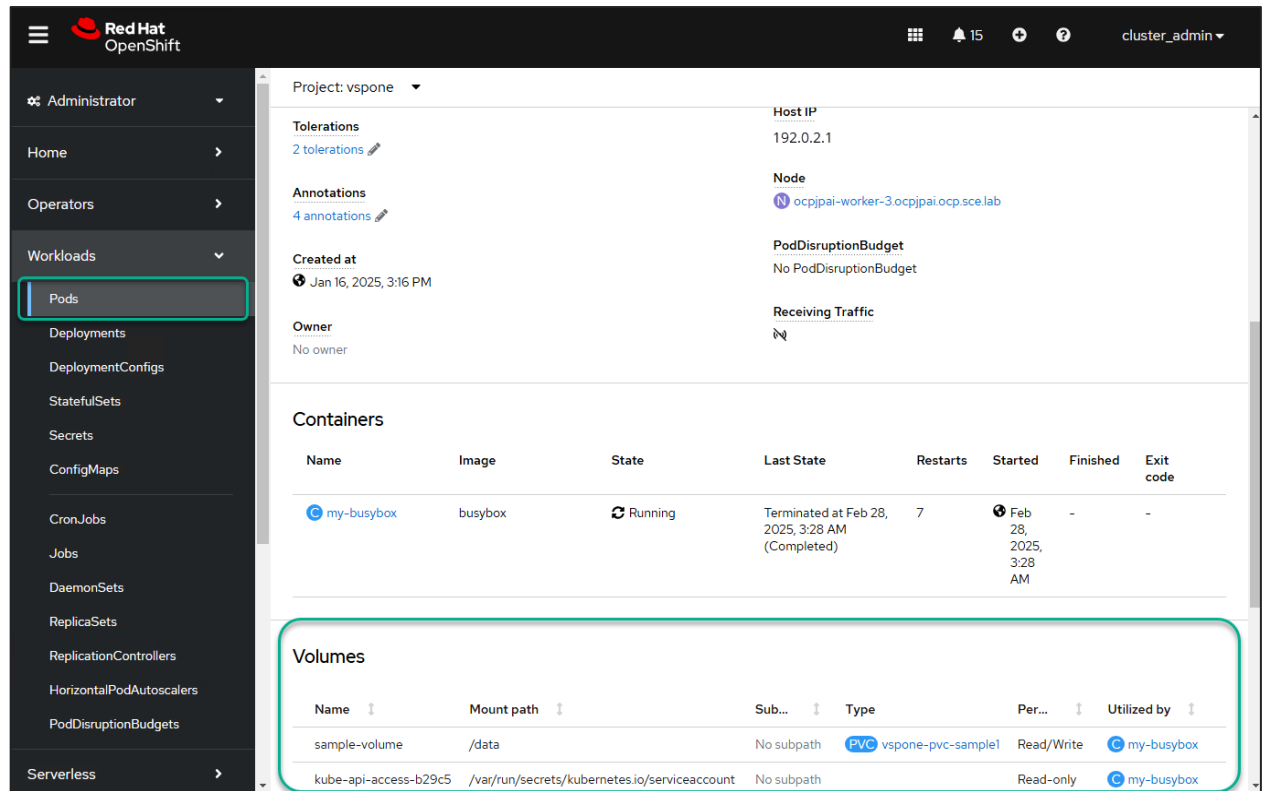
Key	Description
name	Enter a name for the PVC.
storage	Enter the storage capacity for the PVC.
storageClassName	Enter the StorageClass name.

- b. Create the PVC:

```
oc create -f <name-of-the-yaml-file-for-the-PVC>
```

2. Create a pod.

Define the volume details in the pod's Kubernetes resource by specifying the PVC you created. If required, you can define multiple volumes.



a. Create a pod yaml file:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-sample
spec:
  containers:
    - name: my-busybox
      image: busybox
      volumeMounts:
        - mountPath: "/data"
          name: sample-volume
      command: ["sleep", "1000000"]
      imagePullPolicy: IfNotPresent
  volumes:
    - name: sample-volume
      persistentVolumeClaim:
        claimName: pvc-sample
```

Key	Description
name	Enter a name for the pod.
mountPath	Enter the path where the volume is mounted on a container.
claimName	Enter the name of the PVC you created.

b. Create the pod:

```
oc create -f <yaml-file-name-of-the-pod>
```

3. Verify that the volumes are created and mounted.

- In VSP One Block Administrator, verify that the volumes are created in the storage system.

HITACHI VSP One Block Administrator

Monitor
Storage
Servers
Volumes
Pools
Ports
Drives
Replication
Settings

Volumes

Create Volumes

Filter: ID | greater than or equal to | 15 | Volume Type | Is (any one) | Atta...

ALL (2)

	Name	ID	Total	Used	Number of Servers	Pool	Number of	Volume Type
☐	spc-838e66ac1f	15 (0x000F)	10.00 GiB	924.00 MiB	0	hspc-pool	0	Attached Unmanaged
☐	spc-d0852a8891	16 (0x0010)	10.00 GiB	882.00 MiB	0	hspc-pool	0	Attached Unmanaged

Show 25 Rows

- In the Red Hat OpenShift Container Platform web console, verify that the volumes are created and mounted to the pod.

Red Hat OpenShift

You are logged in as a temporary administrative user. Update the cluster OAuth configuration to allow others to log in.

Project: postgresql-vsp-one-block

Containers

Name	Image	State	Last State	Restarts	Started	Finished	Exit code
container-postgres-1	postgres	Running	-	0	Jan 24, 2025, 12:16 AM	-	-

Volumes

Name	Mount path	SubPath	Type	Permissions	Utilized by
postgres-10g-1	/var/lib/postgresql	No subpath	PVC pvc-postgres-1	Read/Write	container-postgres-1
postgres-10g-2	/var/log/postgresql	No subpath	PVC pvc-postgres-2	Read/Write	container-postgres-1
kube-api-access-f26fm	/var/run/secrets/kubernetes.io/serviceaccount	No subpath		Read-only	container-postgres-1

Summary

With HSPC integrated, you can efficiently manage persistent storage in your Kubernetes environment. This lets you deploy stateful applications in containers while ensuring optimized storage performance.

References

- [Storage Plug-in for Containers Quick Reference Guide](#)
- [Virtual Storage Platform One Block Administrator User Guide](#)
- [Virtual Storage Platform One Block System Administrator Guide](#)
- Sample yaml files: <https://github.com/hitachi-vantara/csi-operator-hitachi/blob/main/hspc>